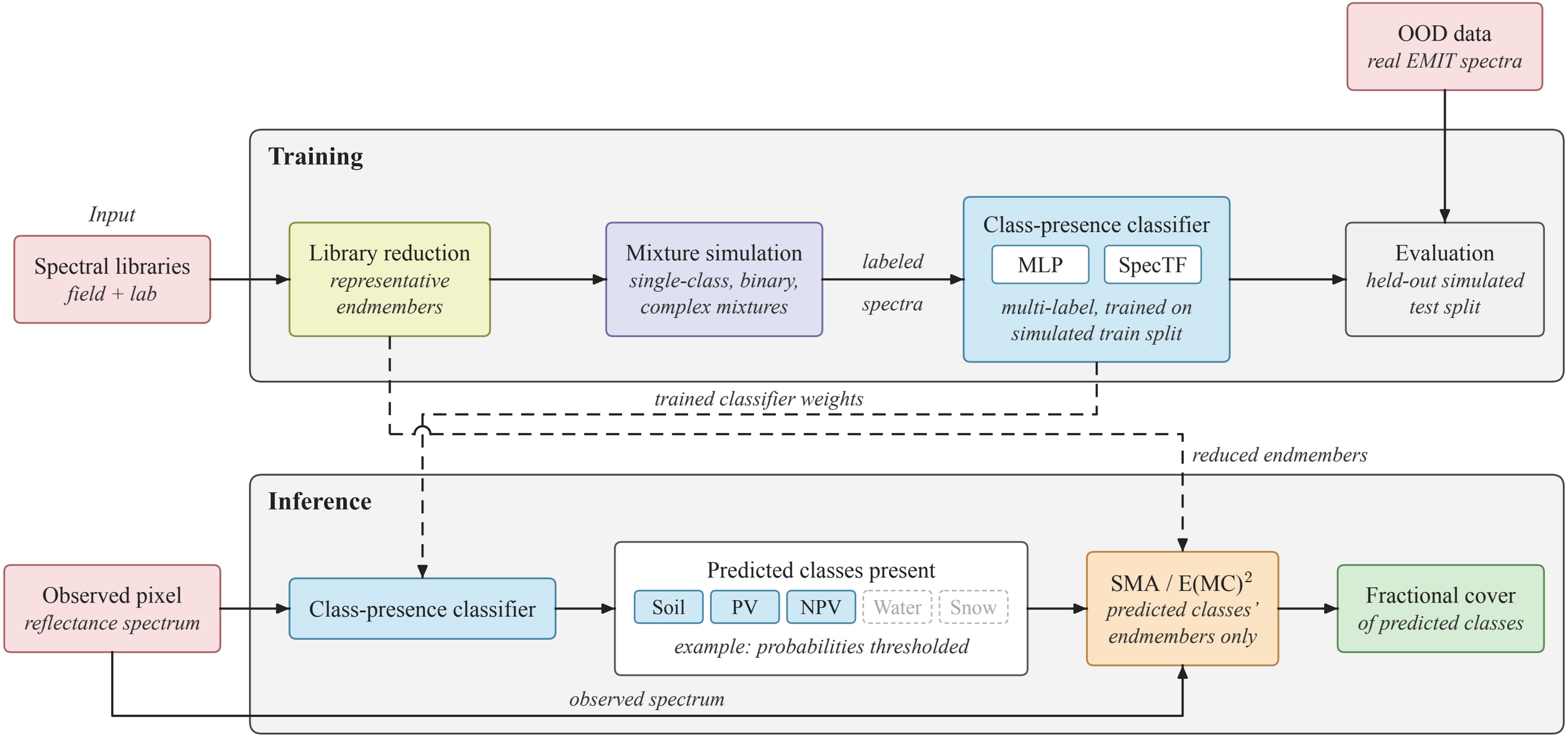

OOD data
real EMIT spectra
Training
Input
Spectral libraries
field + lab
Library reduction
representative endmembers
Mixture simulation
single-class, binary, complex mixtures
labeled spectra
Class-presence classifier
MLP
SpecTF
multi-label, trained on simulated train split
Evaluation
held-out simulated test split
trained classifier weights
reduced endmembers
Inference
Observed pixel
reflectance spectrum
Class-presence classifier
Predicted classes present
Soil
PV
NPV
Water
Snow
example: probabilities thresholded
SMA / E(MC)2
predicted classes' endmembers only
Fractional cover
of predicted classes
observed spectrum